

$$T(cx) = T(cx_1, cx_2, -cx_3) = cT(x_1, x_2, x_3) \quad (\text{Scalar multiplication})$$

$$= c \cdot T(x) \quad \boxed{\therefore \text{Condition 2 satisfied}}$$

• By defn, T is a linear transformation. $\square$

36) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation that projects each vector $x = (x_1, x_2, x_3)$ onto the plane $x_2 = 0$, so $T(x) = (x_1, 0, x_3)$. Show that T is a linear transformation.

Solution: $T(x) = (x_1, 0, x_3) \quad T(y) = (y_1, 0, y_3)$

$$T(x+y) = (x+y_1, 0, x_3+y_3) = (x_1, 0, x_3) + (y_1, 0, y_3) \quad (\text{Vector addition})$$

$$= T(x) + T(y) \quad \boxed{\therefore \text{Condition 1 satisfied}}$$

$$T(cx) = (cx_1, 0, cx_3) = c(x_1, 0, x_3) \quad (\text{Scalar multiplication})$$

$$= c \cdot T(x)$$

$\boxed{\therefore \text{Condition 2 satisfied}}$ • By defn, T is a linear transformation. $\square$